NORTH SOUTH UNIVERSITY · DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING (ECE) · CSE499 CAPSTONE

A Voice-Based AI Framework for Medical Pre-Screening of Bangla & Regional Bangladeshi Speech

with Intelligent Symptom Capture, Risk Assessment, Clinical Documentation & EHR Integration

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1 Introduction

- Patients describe symptoms in **Bangla, Banglish & regional dialects**; before consultation, key details are missed.
- Dialect variation, code-switching & clinic noise make **reliable speech capture hard**.
- We build a **voice-first AI pre-screening** system that captures symptoms before — and never replaces — the doctor.

2 Objective

- Let patients describe symptoms **by voice** in Bangla / Banglish / dialect.
- Extract **structured clinical information** and detect what is missing.
- Ask **targeted follow-up questions** until the case is complete.
- Assess **risk** with safety rules and **explainable** reasoning.
- Produce a doctor-ready record and **integrate with the EHR (FHIR)**.

SAFETY BY DESIGN

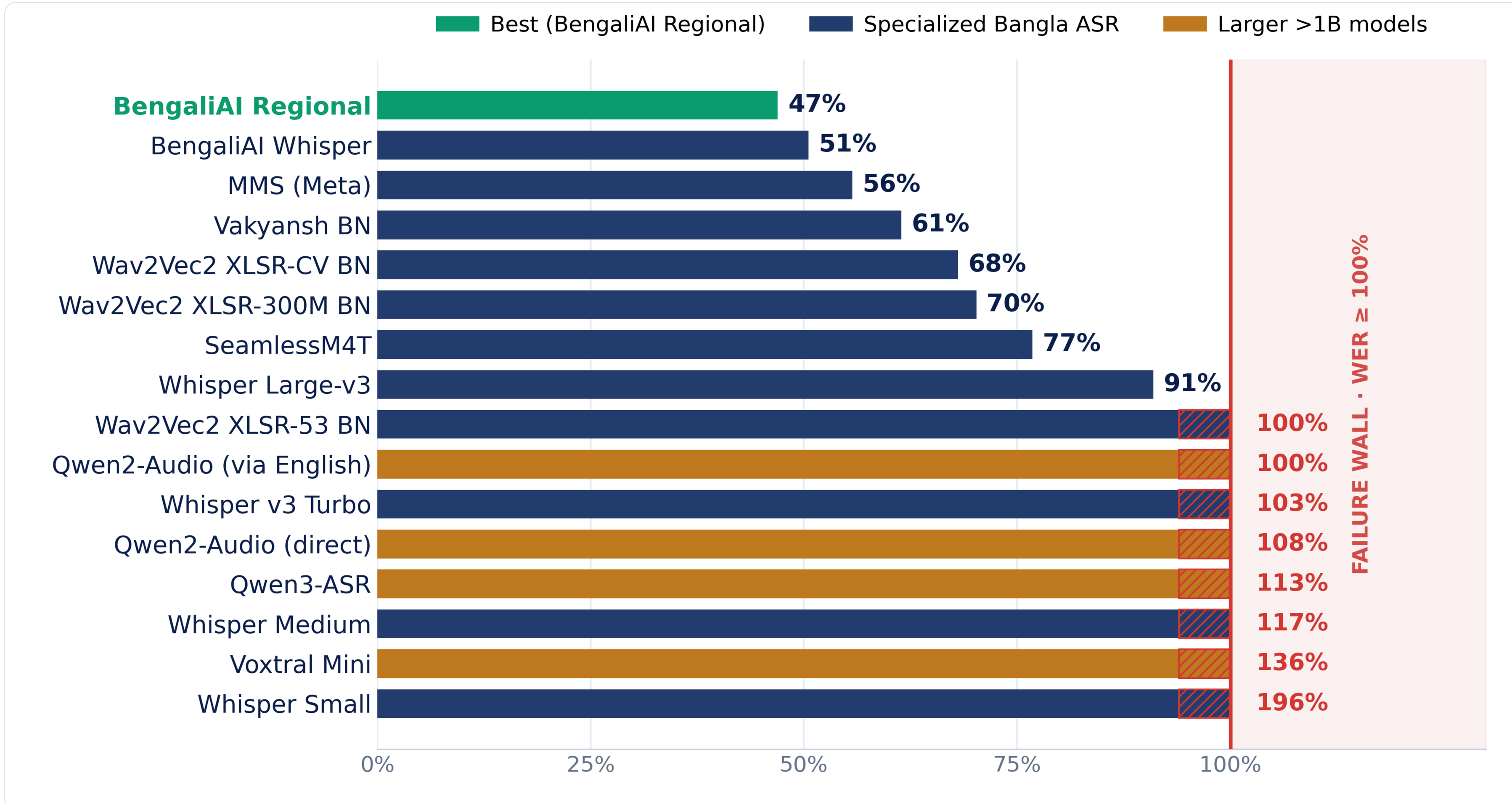
- ✓ Preserve the patient's **exact words**
- ✓ **Never diagnose** — narrows, never names a disease
- ✓ Red-flag symptoms force the **Critical** tier
- ✓ **Synthetic / consented** data only

3 Methodology

1. **SPEAK** — Voice input — Bangla / Banglish / dialect
2. **UNDERSTAND** — Speech → text, cleaned; symptoms extracted — M1–M3
3. **COMPLETE** — Find gaps; follow-up loop until complete — M6–M9 ↺
4. **ASSESS** — Risk Low → Critical + red-flag check; explained — M10–M11
5. **DOCUMENT** — Structured report → EHR; FHIR · PDF · DOCX — M12–M13
6. **REVIEW** — Medic triage & doctor review → consultation — M14–M15

Raw words preserved. 16 ASR systems benchmarked to choose the approach.

4 Results

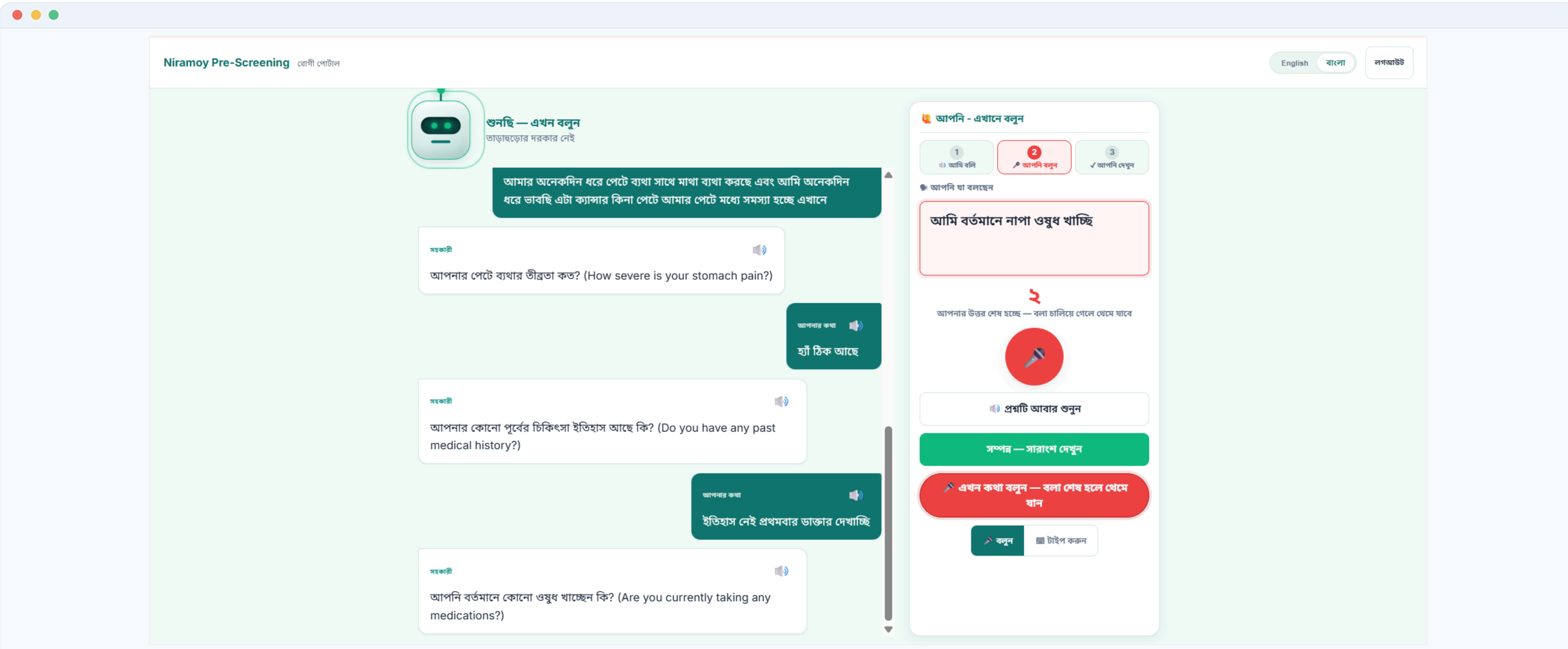


16 ASR systems benchmarked on multi-dialect medical speech: best **47% WER**, **8 failed** (≥100%), larger models worse — proving a single-pass transcript is unsafe.

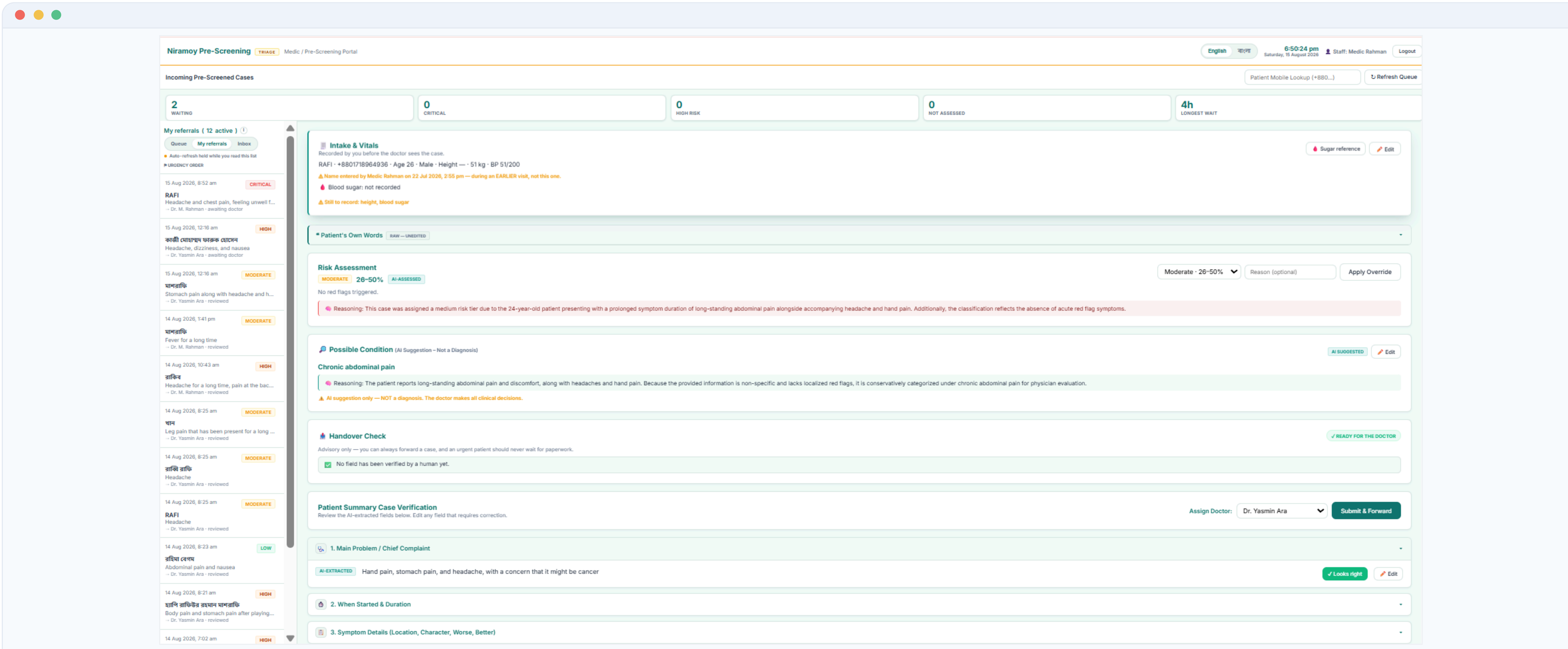
Offline research benchmark — the live prototype uses the browser Web Speech API, not these models.

16	47%	8
models tested	best WER	failed (≥100%)

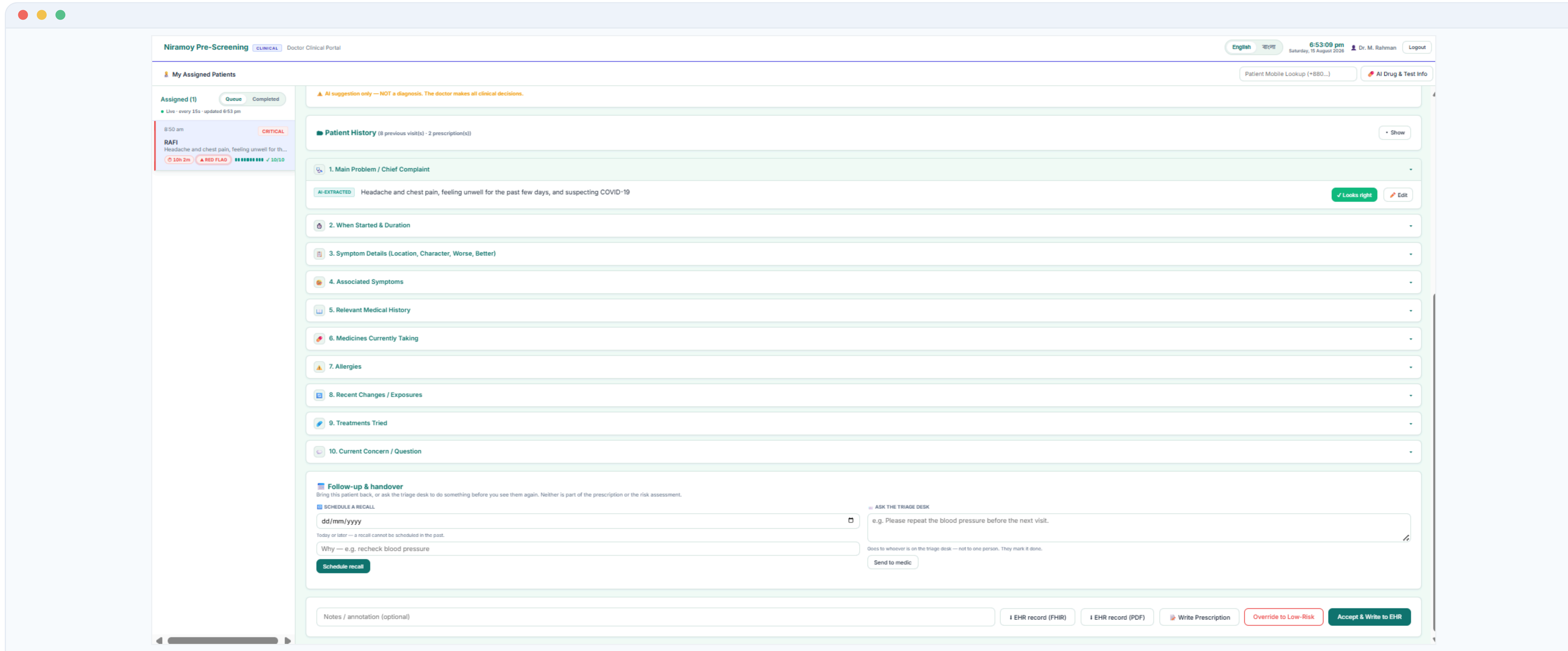
5 System Features — the three portals & the clinical record



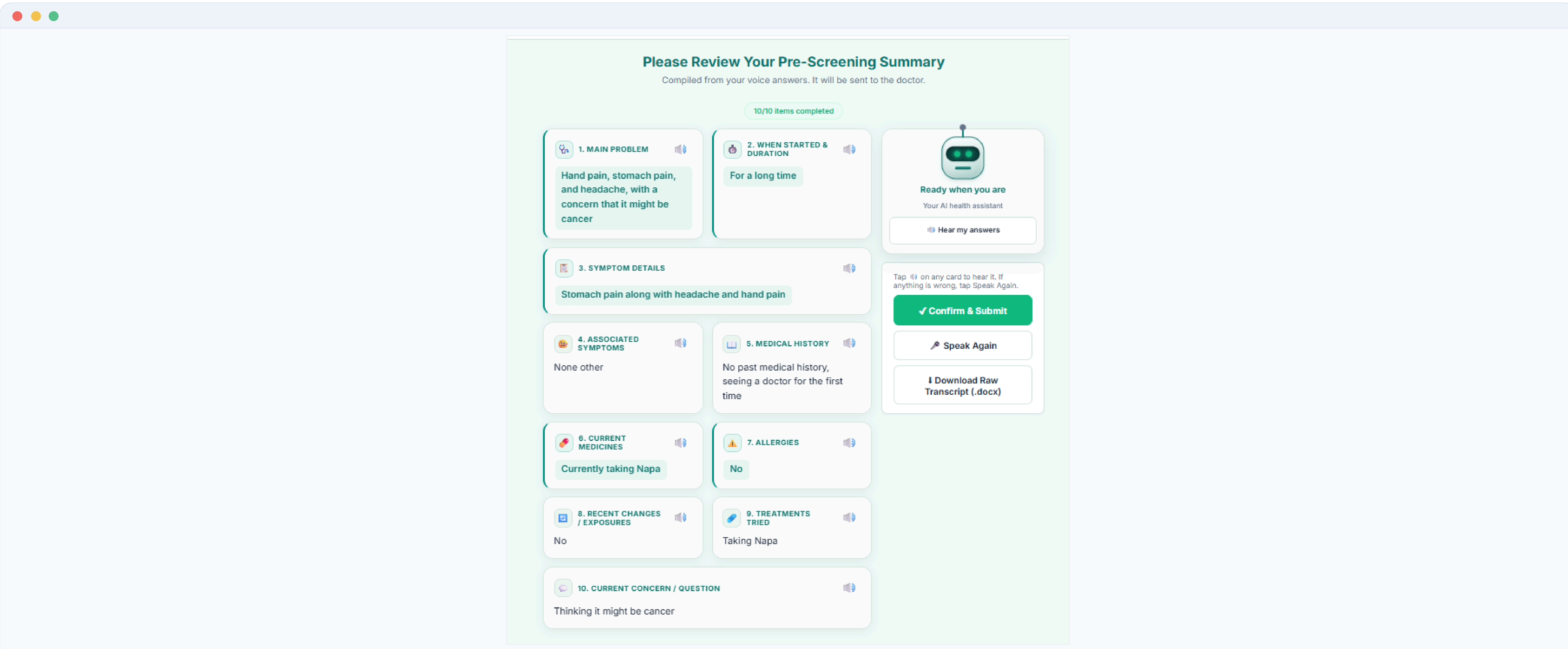
Patient kiosk — live voice conversation & spoken follow-up (Bangla)



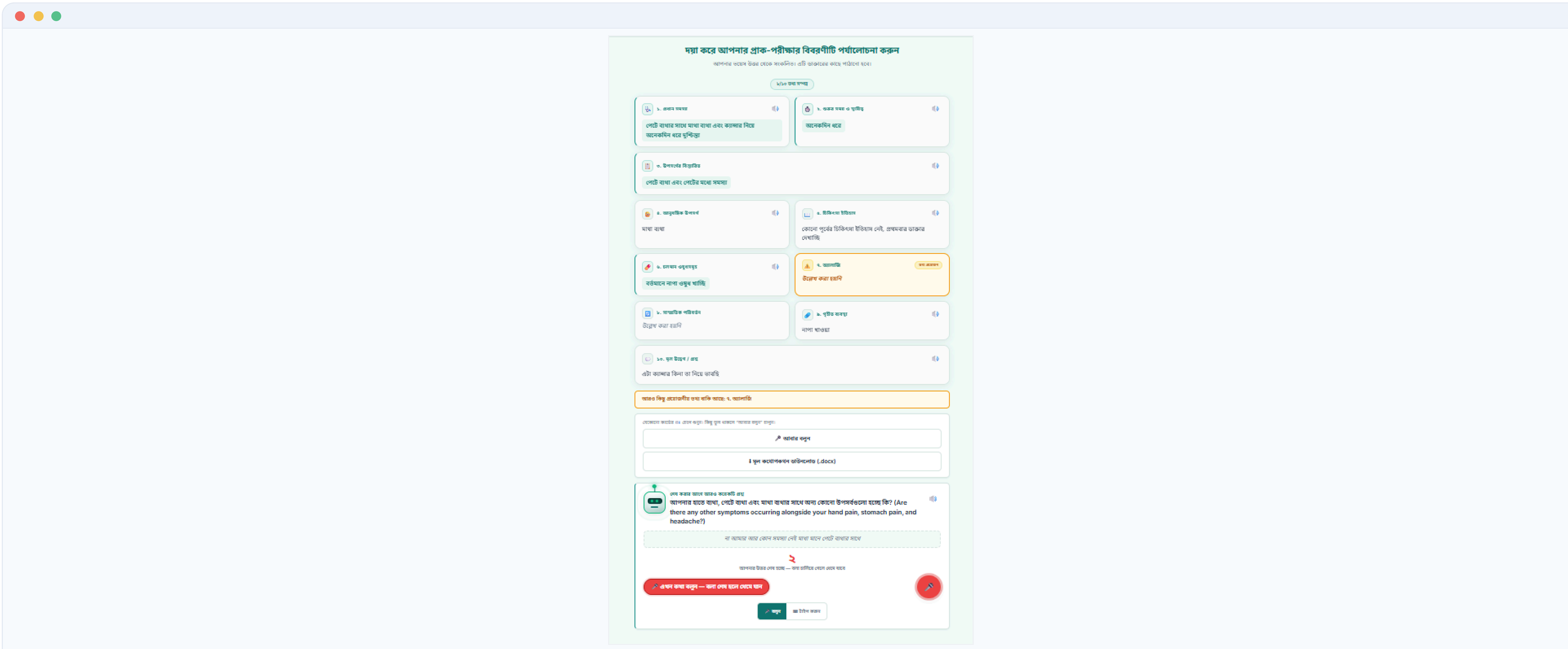
Medic triage — risk-ordered queue, AI reasoning & field verification



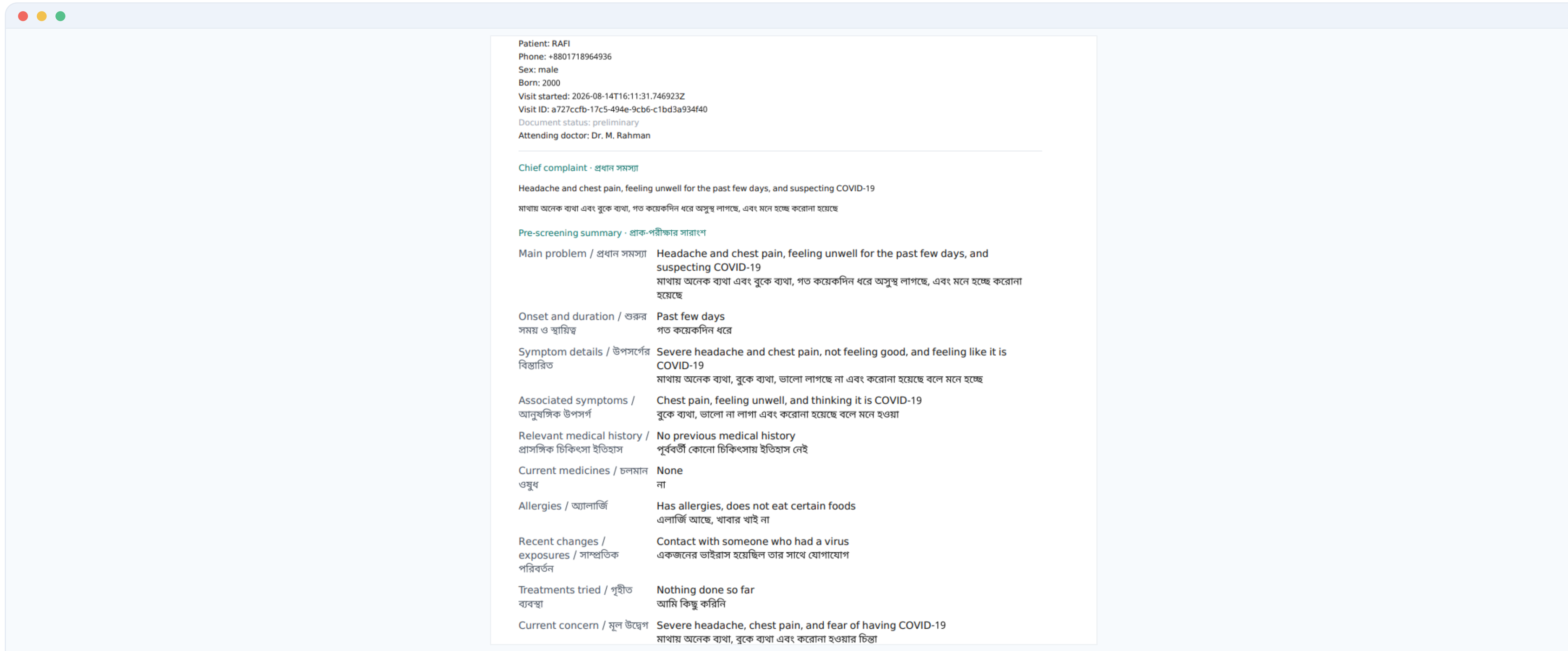
Doctor portal — safety panel, patient history & EHR / prescription actions



Patient review — 10-field structured summary, confirm before submit



Patient review (Bangla) — same summary & follow-up, fully bilingual



Clinical record — structured bilingual EHR document (FHIR · PDF · DOCX)

6 Technology Stack

BACKEND: Python · FastAPI · REST + WebSocket

VOICE: Web Speech API (STT, bn-BD) · edge-tts (Bangla TTS)

AI / LLM: Gemini · Groq · OpenRouter · + fallback

DATA & EXPORT: SQLite + Alembic · HL7 FHIR R4 · fpdf2 · python-docx

FRONTEND: HTML · CSS · JS · Patient · Medic · Doctor portals